# Piyush P. Jirwankar

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## EDUCATION

**University of California, Santa Cruz** 2022 - 2027 (expected)
- Ph.D. in Electrical and Computer Engineering | GPA: 4/4

**Indian Institute of Technology Bombay** 2017 - 2022
- Dual Degree (B.Tech. + M.Tech.) | GPA: 8.93/10
  - B.Tech. in Aerospace Engineering | M.Tech. in Systems and Control Engineering

## SELECTED PUBLICATIONS

[1] **Jirwankar, Piyush P.**, D. E. Ochoa, and R. G. Sanfelice (2026a). "Multiple Control Barrier Functions for Geometric Hybrid Systems: Modeling and Safe Control Synthesis". In: Under review at Springer MCSS.
[2] **Jirwankar, Piyush P.**, D. E. Ochoa, and R. G. Sanfelice (2026b). "Geometric Hybrid Dynamical Systems: Part I-Modeling and Stability". In: *arXiv preprint arXiv:2609.17687*. Under review at IEEE Transactions on Automatic Control.
[3] Ochoa, D. E., **Jirwankar, Piyush P.**, and R. G. Sanfelice (2026). "Solution Sets of Geometric Hybrid Systems: Topological Properties and Existence of Extremal Solutions". In: *Proceedings of the 65th IEEE Conference on Decision and Control (CDC)*.
[4] **Jirwankar, Piyush P.**, R. Montgomery, and R. G. Sanfelice (2026). "Hybrid Control for Robust and Global Obstacle Avoidance Via the Eikonal Equation". In: *Proceedings of the 65th IEEE Conference on Decision and Control (CDC)*.
[5] Basu, H., **Jirwankar, Piyush P.**, R. G. Sanfelice, M. Castroviejo-Fernandez, and I. Kolmanovsky (2026). "A Hybrid Model Predictive Control Framework for Docking and Stabilization of Composite Rigid Spacecraft Dynamics". In: *AIAA SCITECH 2026 Forum*.
[6] **Jirwankar, Piyush P.**, S. Di Cairano, and R. G. Sanfelice (2026). "Input-to-State Stability of Model Predictive Control for Discretized Hybrid Dynamical Systems". In: Under review at the American Control Conference (ACC).
[7] **Jirwankar, Piyush P.**, R. Montgomery, and R. G. Sanfelice (2025). "A Robust and Global Hybrid Complementary Filter on SO(3) using Morse Functions on RP3". In: *2025 American Control Conference (ACC)*, pp. 4473–4478.

## RESEARCH EXPERIENCE

**Ph.D. Researcher @ UC Santa Cruz** Sep 2022 - Present
*Analysis, safe control design, and estimation for hybrid dynamical systems on manifolds*

**Safety Filters for Geometric Hybrid Systems on Manifolds**
- Developed a framework for geometric hybrid systems on abstract manifolds, covering systems with and without inputs
- Extended nonsmooth and set-valued analysis to manifolds, deriving hybrid Lyapunov theorems and invariance principles
- Designed pointwise optimal safety filters for geometric hybrid systems and established conditions for measurable selections

**Model Predictive Control of Hybrid Systems**
- Established conditions for local input-to-state stability of MPC for discretized hybrid systems relative to a compact set
- Designed hybrid MPC with hysteretic switching to model spacecraft collisions while ensuring robust rendezvous and docking

**PDE-based Obstacle Avoidance for Ground Robots and Robotic Manipulators**
- Designed an obstacle-avoidance controller using the Eikonal equation, a Hamilton–Jacobi partial differential equation, to achieve safe, robust, global finite-time stabilization of a compact set for integrator dynamics

**Robust an Global Attitude Estimation for Spacecraft and Quadrotors**
- Developed a hybrid passive complementary filter to overcome the topological obstruction to global attitude estimation
- Achieved robust attitude estimation with faster convergence than the extended Kalman filter (EKF) and multiplicative EKF

**Intern @ MERL: Contact Modeling for Warehouse Automation** Summer 2023
- Built a finite-state machine for rigid-body contact and verified its behavior through a bouncing-ball simulation in Modelica
- Analyzed limit cycles in the bouncing-ball dynamics to provide theoretical validation of the contact model

## MENTORSHIP

**2026 Science Internship Program @ UC Santa Cruz** Summer 2026
*Research mentorship for high school students at UC Santa Cruz*
- Mentored four high school students in implementing and validating Eikonal-based obstacle avoidance on the Quanser QArm
- Taught MATLAB/Simulink, calculus, differential equations, and numerical methods to support the students' robotics research

**Student Satellite Program @ IIT Bombay** 2018 - 2022
*Student-led initiative to advance satellite and space technology at IIT Bombay*
- Led guidance, navigation, and control (GNC); proposed spacecraft rendezvous and docking projects and set progress milestones
- Managed annual student recruitment and mentored new members on guidance, navigation, and control projects

## TECHNICAL SKILLS & COURSES

**Programming** Python, MATLAB/Octave, Simulink, Git/GitHub
**Courses** Optimal Control, State Estimation, Stochastic & Networked Control, Distributed Optimization & Machine Learning